Homework I- Solution

Part A- UNIX

Login as root

1. Display the identity of the current user
   whoami
2. Change to **/home**. Save the calendar of the current month in a new file named **month**
   sudo -i
   cd /home
   cal > month
3. Make the file **month** hidden
   mv month .month
4. Change the permissions on **month** as follow: owner (read, write, execute), group (read and write), other users (no permissions)
   chmod 760 .month
5. Create a new group named **G7**
   groupadd G7
6. Create a new user named **Samer** having **G7** as initial group
   useradd -m -g G7 Samer
7. Assign the password ***pass7*** to **Samer**, and switch to user **Samer**
   passwd Samer
   su Samer
8. In **Samer's** home directory create a new directory named: **Test7**. In **Test7** create a directory **d7**
   mkdir Test7
   mkdir Test7/d7

   Another way

   mkdir Test7/d7 -p

9. In **d7** create the file ***read.txt*** containing the following lines:

   What is Programming?
   Computer Programming is the art of writing computer programs.
   Programming involves activities such as analysis, design, coding and testing,
   Programming is also fun!

   cat > Test7/d7/read.txt

10. Display all lines of **read.txt** that contain the word "Programming" at the beginning of a line

    grep Programming Test7/d7/read.txt

Remark:		
grep ^Programming Test7/d7/read.txt		→ Start with programming
grep Programming Test7/d7/read.txt		→ Contain programming
grep $Programming Test7/d7/read.txt		→ End with programming

11. Display the number of characters in the file **read.txt**
    wc -c Test7/d7/read.txt

12. Copy the file **read.txt** into **/home**. Justify in case of errors
    cp Test7/d7/read.txt /home
    An error appear: no permission for Samer to copy a file to the /home directory

Part B- Choose the Best Answer:

13. **mount -n** is:
    a. Indicates the file system type
    b. mount without writing in /etc/mtab
    c. mount all file systems in /etc/fstab
    d. mount the file system as read/write

14. To display the current **crontab**
    a. crontab -e
    b. crontab -l
    c. crontab -r
    d. crontab -f

crontab -e	Opens the current users' crontab for editing
crontab -l	Lists the current users' crontab entries
crontab -r	Removes the current user's crontab
crontab -f	Invalid no such standard option

UNIX

Login as root

1. Display your working directory

   pwd

2. List the content of the directory **/dev** page by page

   ls /dev | less

3. In **/home** create the directory **Music**; In Music (using one command) create the directories: **Beethoven, Bach**
   mkdir -p /home/Music/{Beethoven,Bach}

   another way
   mkdir /home/Music/Beethoven /home/Music/Bach -p

4. In **Music** create the text files, sonata1, sonata3

   Touch /home/Music/sonata1 /home/Music/sonata3

5. Move all files having a name of 7 characters, starting with "sonata", and the 7th character is the digit 1 or the digit 3 from **Music** to **Bach**

   mv /home/Music/sonata[13] /home/Music/Bach

6. In **Music** create the text file **opera** that contains the lines

   cat > /home/Music/opera

   The symphony No. 9,
   Is Beethoven's final complete master piece.
   This symphony is one of the best-known works in classical music.

7. A. Display all line of the file **opera** containing the word **"symphony"**

   grep "symphony" /home/Music/opera

   B. Display all line of the file **opera** that does not contain the word **"symphony"**

   grep -v "symphony" /home/Music/opera

   If you want to search the lines that start with **symphony** write:
   grep ^symphony /home/Music/opera

   If you want to search the lines that end with **symphony** write:
   grep $symphony /home/Music/opera

8. Create a hard link for the file **opera**
   ln /home/Music/opera /home/Music/opera_link

9. Create a new user account named **Mozart**, having **Classic** as initial group and **/home/Music** as home directory
   sudo groupadd Classic
   sudo useradd -g Classic -d /home/Music Mozart

   If you want to change the password
   sudo passwd Mozart

   If you create the user Akram without defining a group for him, the system create a default group for him (with the same name)
   sudo useradd -d /home/xyz Akram